Exercise Set 10: De Rham Cohomology

In Set 9 we saw that some closed forms cannot be exact and that this failure is detected by integration. De Rham cohomology makes this precise: it is the quotient $H^k_{dR}(M) = \ker d / \operatorname{im} d$, the space of closed k-forms modulo exact ones. This is a topological invariant of M it measures the "holes" of each dimension. The Witten index of a supersymmetric quantum theory living on M computes $\chi(M) = \sum(-1)^k \dim H^k_{dR}(M)$, connecting quantum physics directly to the topology of space.

Recall of Definitions and Examples

De Rham cohomology

R1. Write the definition of the **de Rham cohomology groups** $H^k_{dR}(M)$. Define:

(a) The space of **closed** k-forms: $Z^k(M) = \ker(d : \Omega^k \to \Omega^{k+1})$

(b) The space of **exact** k-forms: $B^k(M) = \operatorname{im}(d : \Omega^{k-1} \to \Omega^k)$

(c) The **k-th de Rham cohomology group**: $H^k_{dR}(M) = Z^k(M)/B^k(M)$

(d) The **cohomology class** $[\omega]$ of a closed form ω

Why is $B^k(M) \subseteq Z^k(M)$? (i.e. why is every exact form closed?)

R2. State the **Poincaré lemma**: for any $k \geq 1$, $H^k_{dR}(\mathbb{R}^n) = 0$. What does this say geometrically? Give an example of a closed form on $\mathbb{R}^n$ and find its potential explicitly.

R3. State the cohomology groups of the following spaces (just the answers you will prove some in the computation section):

(a) $H^k_{dR}(\mathbb{R}^n)$

(b) $H^k_{dR}(S^n)$

(c) $H^k_{dR}(T^n)$ where $T^n = (S^1)^n$

(d) $H^k_{dR}(\mathbb{R}^n \setminus \{0\})$

(e) $H^k_{dR}(\mathbb{R}^2 \setminus \{0\})$

R4. Write the definition of the **Euler characteristic** in terms of de Rham cohomology:

$$\chi(M) = \sum_{k=0}^{\dim M} (-1)^k \dim H^k_{dR}(M).$$

Compute χ for each space in R3.

R5. State the **Künneth formula** for the de Rham cohomology of a product $M \times N$:

$$H^k_{dR}(M \times N) \cong \bigoplus_{p+q=k} H^p_{dR}(M) \otimes H^q_{dR}(N).$$

Use this to compute $H^*_{dR}(T^2)$ from $H^*_{dR}(S^1)$.

Hodge theory

R6. Let (M, g) be a compact oriented Riemannian manifold of dimension n. The **Hodge star** $* : \Omega^k(M) \to \Omega^{n-k}(M)$ is defined by:

$$\alpha \wedge *\beta = \langle \alpha, \beta \rangle \operatorname{vol}_g$$

for $\alpha, \beta \in \Omega^k(M)$, where $\langle \cdot, \cdot \rangle$ is the pointwise inner product induced by g.

Write the action of $*$ on the basis $\{dx^i\}$ for $\mathbb{R}^3$:

(a) $*1 = ?$, $*(dx \wedge dy \wedge dz) = ?$

(b) $*dx = ?$, $*dy = ?$, $*dz = ?$

(c) $*(dx \wedge dy) = ?$, $*(dy \wedge dz) = ?$, $*(dz \wedge dx) = ?$

R7. Write the definition of the **codifferential** $d^* : \Omega^k(M) \to \Omega^{k-1}(M)$:

$$d^* = (-1)^{n(k+1)+1} * d * .$$

What is $(d^*)^2$? What is d^* on $\Omega^0(M)$?

R8. Write the definition of the **Hodge Laplacian** $\Delta = dd^* + d^*d : \Omega^k(M) \to \Omega^k(M)$. A form ω is **harmonic** if $\Delta\omega = 0$. State the **Hodge theorem**: on a compact oriented Riemannian manifold,

$$H^k_{dR}(M) \cong \mathcal{H}^k(M) := \ker \Delta|_{\Omega^k(M)}.$$

What does this say? (Every cohomology class has a unique harmonic representative.)

Theory

T1. (De Rham cohomology is a topological invariant.) Show that if $f : M \to N$ is a smooth map, the pullback f^* induces a well-defined linear map on cohomology:

$$f^* : H^k_{dR}(N) \to H^k_{dR}(M).$$

That is: show f^* maps closed forms to closed forms and exact forms to exact forms. Conclude: if f is a diffeomorphism, f^* is an isomorphism of cohomology groups.

T2. (Homotopy invariance.) Two smooth maps $f, g : M \to N$ are **smoothly homotopic** if there exists a smooth map $H : M \times [0,1] \to N$ with $H(\cdot, 0) = f$ and $H(\cdot, 1) = g$. Show:

$$f^* = g^* : H^k_{dR}(N) \to H^k_{dR}(M).$$

Hint: construct a chain homotopy operator $K : \Omega^k(N) \to \Omega^{k-1}(M)$ such that $f^* - g^* = dK + Kd$.

Conclude: if M and N are homotopy equivalent (there exist maps $f : M \to N$ and $g : N \to M$ with $g \circ f \simeq \mathrm{id}_M$ and $f \circ g \simeq \mathrm{id}_N$), then $H^*_{dR}(M) \cong H^*_{dR}(N)$.

T3. (Poincaré lemma from homotopy invariance.) $\mathbb{R}^n$ is contractible it is homotopy equivalent to a point.

(a) Show $H^k_{dR}(\{\mathrm{pt}\}) = 0$ for $k \geq 1$.

(b) Use homotopy invariance to conclude $H^k_{dR}(\mathbb{R}^n) = 0$ for $k \geq 1$.

T4. (Mayer-Vietoris sequence.) Let $M = U \cup V$ where $U, V \subseteq M$ are open. The **Mayer-Vietoris sequence** is the long exact sequence:

$$\cdots \to H^k_{dR}(M) \xrightarrow{(i^*, j^*)} H^k_{dR}(U) \oplus H^k_{dR}(V) \xrightarrow{k^* - \ell^*} H^k_{dR}(U \cap V) \xrightarrow{\delta} H^{k+1}_{dR}(M) \to \cdots$$

Use Mayer-Vietoris to compute $H^*_{dR}(S^n)$ by cutting S^n into two hemispheres U and V with $U \cap V \simeq S^{n-1}$.

Computation

Computing cohomology groups

C1. ($H^*_{dR}(S^1)$.)

(a) What are $\Omega^0(S^1)$ and $\Omega^1(S^1)$? Write their local forms in the angle chart θ.

(b) A closed 0-form satisfies $df = 0$. What are all closed 0-forms on S^1? What are the exact 0-forms? Compute $H^0_{dR}(S^1)$.

(c) Every 1-form on S^1 is closed (why?). Which 1-forms are exact? Show that $\omega = f(\theta)\, d\theta$ is exact iff $\int_{S^1} \omega = 0$. Conclude $H^1_{dR}(S^1) \cong \mathbb{R}$, generated by $[d\theta]$.

(d) Verify: $\chi(S^1) = \dim H^0 - \dim H^1 = 1 - 1 = 0$. ✓

C2. ($H^*_{dR}(S^2)$.)

(a) $H^0_{dR}(S^2) = ?$ (connected space)

(b) Every 1-form on S^2 satisfies: show $H^1_{dR}(S^2) = 0$. *Hint: use Mayer-Vietoris with $U = S^2 \setminus \{N\}$ and $V = S^2 \setminus \{S\}$, both contractible.*

(c) Show $H^2_{dR}(S^2) \cong \mathbb{R}$, generated by $[\sin\theta\, d\theta \wedge d\varphi]$. *Hint: if ω is exact, $\int_{S^2} \omega = 0$. But $\int_{S^2} \sin\theta\, d\theta \wedge d\varphi = 4\pi \neq 0$.*

(d) Verify $\chi(S^2) = 1 - 0 + 1 = 2$. ✓

C3. ($H^*_{dR}(T^2)$ via Künneth Nawata HW3 Problem 4.)

Using $H^0_{dR}(S^1) = H^1_{dR}(S^1) = \mathbb{R}$ and the Künneth formula:

$$H^k_{dR}(T^2) = H^k_{dR}(S^1 \times S^1) \cong \bigoplus_{p+q=k} H^p_{dR}(S^1) \otimes H^q_{dR}(S^1).$$

(a) Compute $H^0_{dR}(T^2)$, $H^1_{dR}(T^2)$, $H^2_{dR}(T^2)$.

(b) Write the explicit generators of each cohomology group as differential forms in coordinates (θ_1, θ_2) on T^2.

(c) Verify $\chi(T^2) = 1 - 2 + 1 = 0$. ✓

(d) Describe the geometric meaning of the two generators of $H^1_{dR}(T^2) \cong \mathbb{R}^2$: what do $[d\theta_1]$ and $[d\theta_2]$ detect?

C4. ($H^*_{dR}(\mathbb{R}^2 \setminus \{0\})$.)

(a) Show $\mathbb{R}^2 \setminus \{0\}$ is homotopy equivalent to S^1. Write the explicit homotopy.

(b) Conclude $H^k_{dR}(\mathbb{R}^2 \setminus \{0\}) \cong H^k_{dR}(S^1)$.

(c) The generator of $H^1_{dR}(\mathbb{R}^2 \setminus \{0\})$ is the angle form $\omega = \frac{-y\,dx + x\,dy}{x^2+y^2}$. Verify $d\omega = 0$ and $\oint_{S^1} \omega = 2\pi \neq 0$.

(d) Connect to Set 4: the winding number of a curve γ around the origin is $\frac{1}{2\pi} \oint_\gamma \omega$. The cohomology class $[\omega] \in H^1_{dR}(\mathbb{R}^2 \setminus \{0\})$ is what the winding number measures.

C5. (Harmonic forms and the Hodge theorem on S^1.)

(a) The Riemannian metric on S^1 is $g = d\theta^2$. The Hodge star acts by $*1 = d\theta$ and $*d\theta = 1$. Compute the codifferential $d^* = - * d *$ acting on $\Omega^1(S^1)$: for $\omega = f(\theta)\, d\theta$, show $d^*\omega = -f'(\theta)$.

(b) The Hodge Laplacian on $\Omega^0(S^1)$ is $\Delta f = d^* df = -f''(\theta)$. Find all harmonic functions ($\Delta f = 0$) on S^1. What is $\mathcal{H}^0(S^1)$?

(c) The Hodge Laplacian on $\Omega^1(S^1)$ is $\Delta(f\, d\theta) = -f''(\theta)\, d\theta$. Find all harmonic 1-forms on S^1. What is $\mathcal{H}^1(S^1)$?

(d) Verify: $\mathcal{H}^0(S^1) \cong H^0_{dR}(S^1) \cong \mathbb{R}$ and $\mathcal{H}^1(S^1) \cong H^1_{dR}(S^1) \cong \mathbb{R}$. The unique harmonic representative of $[d\theta]$ is $d\theta$ itself.

C6. (The generator of $H^n_{dR}(S^n)$ Nawata HW3 Problem 3.)

(a) Recall from Set 9 (P1): the form $\iota^*\omega$ on S^n is closed and satisfies $\int_{S^n} \iota^*\omega = \mathrm{Vol}(S^n) \neq 0$. Conclude $[\iota^*\omega]$ is a nonzero class in $H^n_{dR}(S^n)$.

(b) Since $H^n_{dR}(S^n) \cong \mathbb{R}$, the class $[\iota^*\omega]$ generates the entire group. Write this generator explicitly for $n = 1$ and $n = 2$.

(c) The Hodge theorem says this generator has a unique harmonic representative. For S^2 with the round metric, show that $\sin\theta\, d\theta \wedge d\varphi$ is harmonic: compute $\Delta(\sin\theta\, d\theta \wedge d\varphi) = 0$.

Physics: Supersymmetric Quantum Mechanics and the Witten Index

Witten (1982) showed that de Rham cohomology can be recovered from a quantum mechanical system living on M. The Hilbert space is the space of all differential forms $\Omega^*(M)$, the supercharges are $Q = d$ and $Q^\dagger = d^*$, and the Hamiltonian is the Hodge Laplacian $H = \Delta$. The ground states are harmonic forms exactly the de Rham cohomology. The Witten index $\mathrm{Tr}[(-1)^F e^{-\beta H}]$ computes $\chi(M)$, independent of the "temperature" $1/\beta$. This is the first rigorous example of a topological quantum field theory.

P1. (Building the supersymmetric quantum mechanics.)

The **Hilbert space** is $\mathcal{H} = \Omega^*(M) = \bigoplus_{k=0}^n \Omega^k(M)$, the space of all differential forms on M. The **fermion number operator** $(-1)^F$ acts as $+1$ on even-degree forms and -1 on odd-degree forms. The **supercharges** are:

$$Q = d : \Omega^k \to \Omega^{k+1}, \qquad Q^\dagger = d^* : \Omega^k \to \Omega^{k-1}.$$

(a) Show that $Q^2 = 0$ and $(Q^\dagger)^2 = 0$. Which property of d gives $Q^2 = 0$?

(b) The **Hamiltonian** is the anticommutator:

$$H = \{Q, Q^\dagger\} = QQ^\dagger + Q^\dagger Q = dd^* + d^*d = \Delta.$$

Show $[H, Q] = 0$ and $[H, Q^\dagger] = 0$, i.e. Q and $Q^\dagger$ commute with H.
(This says the supercharges are conserved they commute with the Hamiltonian.)

(c) A state ω is a **ground state** if $H\omega = 0$, i.e. $\Delta\omega = 0$. Show that $H\omega = 0$ iff $Q\omega = 0$ and $Q^\dagger\omega = 0$, i.e. iff $d\omega = 0$ and $d^*\omega = 0$.
Hint: use $\langle H\omega, \omega\rangle = \|Q\omega\|^2 + \|Q^\dagger\omega\|^2 \geq 0$.

(d) By the Hodge theorem, the ground states in degree k are $\mathcal{H}^k(M) \cong H^k_{dR}(M)$. Write this as a table for $M = T^2$:

Degree k	Ground states $\mathcal{H}^k(T^2)$	$\dim H^k_{dR}(T^2)$
0	constants	
1	$a\, d\theta_1 + b\, d\theta_2$	
2	$c\, d\theta_1 \wedge d\theta_2$	

(e) **Supersymmetric pairing.** For any non-ground state ω with $H\omega = E\omega$ ($E > 0$): show $Q\omega$ is also an eigenstate with energy E but degree $k+1$, and $Q^\dagger\omega$ has energy E and degree $k-1$. Non-ground states come in **superpartner pairs**. Draw a diagram showing how Q and $Q^\dagger$ pair up states.

P2. (The Witten index.)

The **partition function** of the SQFT is:

$$Z(\beta) = \mathrm{Tr}_{\mathcal{H}}\left[(-1)^F e^{-\beta H}\right] = \sum_{k=0}^{n}(-1)^k \sum_{\lambda\geq 0} e^{-\beta\lambda}\cdot(\text{number of } k\text{-forms with eigenvalue } \lambda).$$

(a) Show that non-ground states ($\lambda > 0$) contribute zero to $Z(\beta)$ because they come in superpartner pairs with opposite sign under $(-1)^F$.
Hint: if ω is a k-form eigenstate with $E > 0$, then $Q\omega$ is a $(k+1)$-form eigenstate with the same E. Their contributions to $Z(\beta)$ are $(-1)^k e^{-\beta E}$ and $(-1)^{k+1}e^{-\beta E}$. These cancel.

(b) Conclude:

$$Z(\beta) = \sum_{k=0}^{n}(-1)^k \dim \mathcal{H}^k(M) = \sum_{k=0}^{n}(-1)^k \dim H^k_{dR}(M) = \chi(M).$$

The Witten index is **independent of** β and equals $\chi(M)$.

(c) Compute $Z(\beta)$ for each of the following by computing $\chi(M)$:

(i) $M = S^1$: $H^0 = H^1 = \mathbb{R}$

(ii) $M = S^2$: $H^0 = H^2 = \mathbb{R}$, $H^1 = 0$

(iii) $M = T^2$: $H^0 = H^2 = \mathbb{R}$, $H^1 = \mathbb{R}^2$

(iv) $M = \Sigma_g$ (genus g surface): $H^0 = H^2 = \mathbb{R}$, $H^1 = \mathbb{R}^{2g}$

(d) **Why the index is topological.** The Witten index is independent of β (the inverse temperature) and of the Riemannian metric g on M (which determines $H = \Delta$). This means:

- Deforming the metric continuously cannot change $Z(\beta)$
- Turning on a "superpotential" (Morse function $s : M \to \mathbb{R}$) cannot change $Z(\beta)$
- Only the topology of M determines $Z(\beta) = \chi(M)$

Connect this to Set 6 Part B: we computed $Z = \chi(M)$ there using Morse theory localization. Now we see it is the same as counting harmonic forms. Write one sentence explaining why both computations must agree.

P3. (Explicit spectrum on S^1.)

On S^1 with the round metric $g = d\theta^2$ and circumference 2π, the eigenforms of Δ are:

$$\Omega^0: \quad f_n(\theta) = e^{in\theta}, \quad \Delta f_n = n^2 f_n \qquad (n \in \mathbb{Z})$$

$$\Omega^1: \quad \omega_n = e^{in\theta}\, d\theta, \quad \Delta\omega_n = n^2\omega_n \qquad (n \in \mathbb{Z})$$

(a) What are the ground states ($n = 0$)? Verify $\mathcal{H}^0(S^1) = \{\text{constants}\}$ and $\mathcal{H}^1(S^1) = \{c\, d\theta\}$.

(b) Compute $Z(\beta)$ explicitly using the full spectrum:

$$Z(\beta) = \sum_{n\in\mathbb{Z}} e^{-\beta n^2} \cdot (+1) - \sum_{n\in\mathbb{Z}} e^{-\beta n^2} \cdot (-1) \cdot (-1).$$

Wait does this give 0? Check: the $n = 0$ terms give $1 - 1 = 0$, and for $n \neq 0$ the pairs cancel. So $Z(\beta) = 0$.

(c) Verify $Z(\beta) = \chi(S^1) = 0$. ✓

(d) **The superpartner pairing on S^1:** for each $n \neq 0$, the pair (f_n, ω_n) are superpartners related by $Qf_n = \omega_n$. Draw the spectrum showing the pairing.

P4. (Connection to the running theme: physics detecting topology.)

Collect the field theories we have used to compute topological invariants:

Set	Field theory	Fields	Invariant	What it detects
6B	1D SQFT (Morse)	x, ρ, χ	$\chi(M)$	critical points
6B	Gauss-Bonnet	metric g	$\chi(\Sigma_g)$	genus (2D)
9	Integration	forms ω	$\int_M \omega$	holes / cycles
10	SQFT (Witten)	forms ω	$\chi(M)$	ground states

(a) In Set 6B (Morse theory), we computed $Z = \sum_i \operatorname{sign}(s'(x_i))$ by localization at critical points. In this set, we computed $Z = \chi(M)$ by counting harmonic forms. The **Morse inequalities** connect these: $\dim H^k_{dR}(M) \leq M_k$ where M_k is the number of index-k critical points of s. Show that $\sum_k (-1)^k \dim H^k_{dR}(M) = \sum_k (-1)^k M_k = Z$ both give the same Euler characteristic.

(b) The Witten index is β-independent: $Z(\beta) = Z(\infty) = Z(0)$. At $\beta \to \infty$ (low temperature): only ground states contribute, $Z = \chi(M)$. At $\beta \to 0$ (high temperature): all states contribute, but superpartner cancellation still gives $Z = \chi(M)$. In the Morse theory computation (Set 6B), which limit corresponds to taking $t \to \infty$?

(c) **The big picture.** Every TQFT we have studied computes a topological invariant by having a partition function that is independent of some continuous parameter (t, β, the metric). This independence is always guaranteed by a symmetry of the action (BRST, supersymmetry, diffeomorphism invariance). Write the common pattern:

$$\text{symmetry} \implies \frac{d}{d(\text{parameter})} Z = 0 \implies Z = \text{topological invariant}.$$

This is physics at its most powerful: a quantum mechanical system built from the geometry of M computes a topological invariant of M. The symmetry (supersymmetry) forces the answer to be topological.

In the next sets we introduce Lie groups (the continuous symmetry groups of physics) and fiber bundles (the geometric setting for gauge theories). These will eventually lead to Chern-Weil theory, where characteristic classes the higher-dimensional analogues of the Euler class are computed as integrals of curvature forms. That is the full story: curvature (local, geometric) computes topology (global).***Reflection.*** *De Rham cohomology* $H^k_{dR}(M) = \ker d/\operatorname{im} d$ *measures the* k*-dimensional holes of* M *the ways that closed forms fail to be exact. The Hodge theorem identifies each cohomology class with a unique harmonic form, the ground state of a quantum system. The Witten index* $Z = \mathrm{Tr}[(-1)^F e^{-\beta H}] = \chi(M)$ *is independent of temperature because superpartner pairs cancel only the ground states (harmonic forms) contribute.*

This is physics at its most powerful: a quantum mechanical system built from the geometry of M *computes a topological invariant of* M*. The symmetry (supersymmetry) forces the answer to be topological.*

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